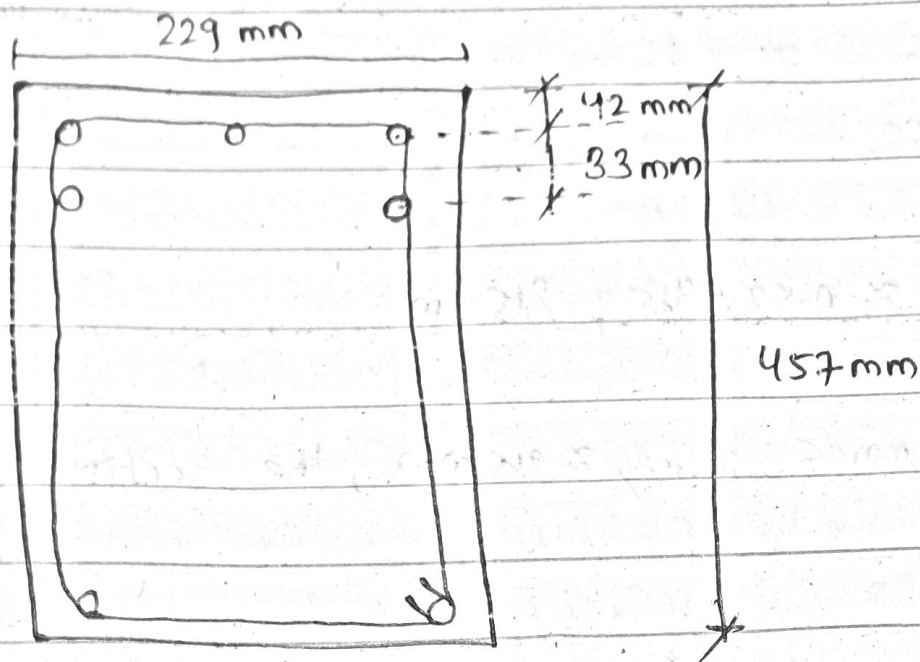


Final Project: Hand Calculations

Beam Section,



7-D16 Rebars

Beam width = $b = 229 \text{ mm}$

Beam Depth = $D = 457 \text{ mm}$

Effective Cover = $d' = 42 \text{ mm}$

Effective Depth = $d = 457 - 42 = 415 \text{ mm}$

$f_{cr} = 39 \text{ MPa}$

$f_y = 295 \text{ MPa}$

Area of rebar = $\frac{\pi}{4} \times 16^2 = 201.06 \text{ mm}^2$

① For Sagging Moment (Bottom steel in Tension)

$d = 415 \text{ mm}$

$A_{st} = 2 \times 201.06 = 402.12 \text{ mm}^2$

Using IS 456-2000,
from Annex-G,

$$\text{Depth of neutral axis } (x_u) = \frac{0.87 f_y A_{st}}{0.36 f_{cr} b} = \frac{0.87 \times 295 \times 402.12}{0.36 \times 39 \times 229} = 32.1 \text{ mm}$$

2

from clause 38.1,
for $f_y = 295 \text{ MPa}$,

$$\frac{x_{u, \max}}{d} \approx 0.52$$

$$\Rightarrow x_{u, \max} \approx 0.52 \times 485 = 252 \text{ mm}$$

Since $x_u = 32.10 \text{ mm} < x_{u, \max} \approx 252 \text{ mm}$, the section is underreinforced,

$$\therefore \text{Design Moment } (M_u^+) = 0.87 f_y A_{st} (d - 0.416 x_u)$$

$$= 0.87 \times 295 \times 402.12 (485 - 0.416 \times 32.1)$$

$$= 46.45 \text{ kN-m}$$

And,

$$\text{Nominal Moment Capacity } (M_{u, \text{nominal}}^+) = f_y A_{st} (d - 0.416 x_u)$$

$$= 47.64 \text{ kN-m}$$

② For Hogging Moment (Top Steel in Tension)

$$x_u = \frac{0.87 \times 295 \times 5 \times 201.06}{0.36 \times 39 \times 229} = 80.25 \text{ mm}$$

$$d = \frac{3 \times 485 + 2 \times (485 - 33)}{5} = 401.80 \text{ mm}$$

$$\text{Design Moment } (M_u^+) = 0.87 \times 295 \times 5 \times 201.06 \left(401.80 - 0.416 \times 80.25 \right)$$

$$= 95.06 \text{ kNm}$$

And,

$$\text{Nominal Moment Capacity } (M_{\text{nominal}}) = 295 \times 5 \times 201.06 \left(401.80 - 0.416 \times 80.15 \right) \\ = 113.77 \text{ kNm} \\ = 109.26 \text{ kNm}$$

(iii) Yield curvature,

$$\text{Modular ratio } (m) = \frac{E_s}{E_c} = \frac{210 \times 10^9}{36 \times 10^9} = 5.833$$

Actual Depth of Neutral axis (n)

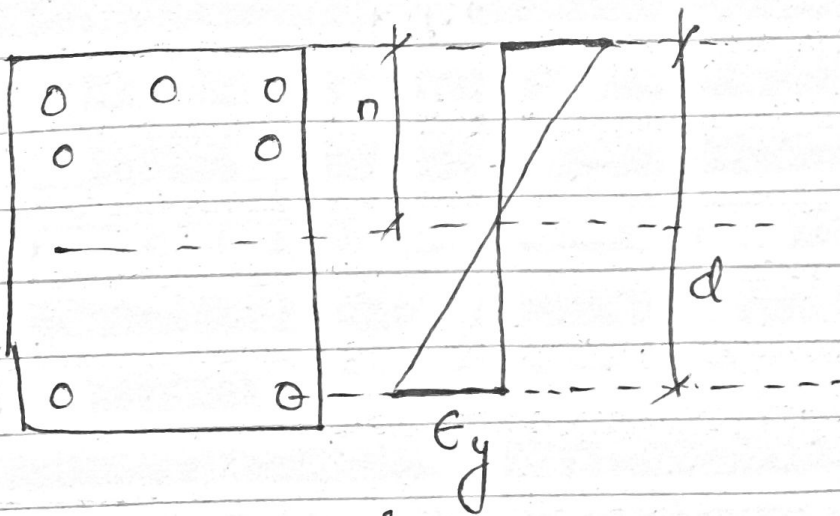
$$(b \times n) \times \frac{n}{2} = m \times A_{st} \times (d - n)$$

$$\Rightarrow \frac{229 \times n^2}{2} = 5.833 \times 402.12 \times (415 - n)$$

$$\Rightarrow n = 82.53 \text{ mm}$$

$$\text{yield strain } (\epsilon_y) = \frac{f_y}{E_s} = \frac{295}{210 \times 10^3} = 1.405 \times 10^{-3}$$

for linear strain distribution,



$$\phi_y \approx \frac{\epsilon_y}{d - n} = \frac{1.404 \times 10^{-3}}{(415 - 82.53) \times 10^{-3}} = 4.23 \times 10^{-3} \text{ m}^{-1}$$

4

(IV) For the cantilever,

$$\text{Total length} = 1916 + 203 = \text{flexible part} + \text{rigid part}$$

$$= 2119 \text{ mm}$$

$$= 2.119 \text{ m}$$

$$\text{Moment at fix end} = \text{Tip load} \times 2.119$$

$$= P \times 2.119$$

$$= 2.119P \quad \text{--- (a)}$$

for Sagging Moment of 47.64 kNm,

$$P_+ = \frac{47.64}{2.119} = 22.48 \text{ kN}$$

for Hogging Moment of ~~113.17 kNm~~, 109.26 kNm,

$$P_- = \frac{113.17}{2.119} = 53.407 \text{ kN}$$

$$P_- = \frac{109.26}{2.119} = 51.56 \text{ kN}$$

Assignment:

Start with moment-curvature analysis. Build a section model with unsymmetric reinforcement and get the moment curvature response. Compare with what you have for the project and see if it makes sense. Also, compare with hand calculations.

Then, build a cantilever beam model with unsymmetric section reinforcement. Get the force-displacement response. Compare with hand calculations.

Comparison with Hand Calculations: Beam Section

This report validates the OpenSees numerical models against manual calculations conducted in accordance with IS 456:2000.

Codes:

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/steps_back/Moment_Curvature_Monotonic.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/steps_back/Moment_Curvature_Cyclic.py

https://github.com/nirajkyadav/Homeworks/blob/main/Final%20Project/steps_back/Cantilever.py

Beam Section A-A

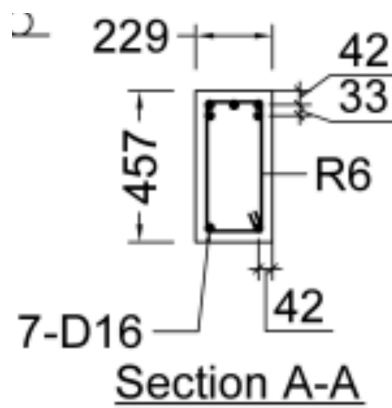
5-D16 top bars + 2-D16 bottom bars

Dimension: 229 × 457 mm

Cover: 42 mm

Concrete: $f_c = 39$ MPa, $E = 36$ GPa

Steel: $f_y = 295$ MPa, $E = 210$ GPa



Numerical Results

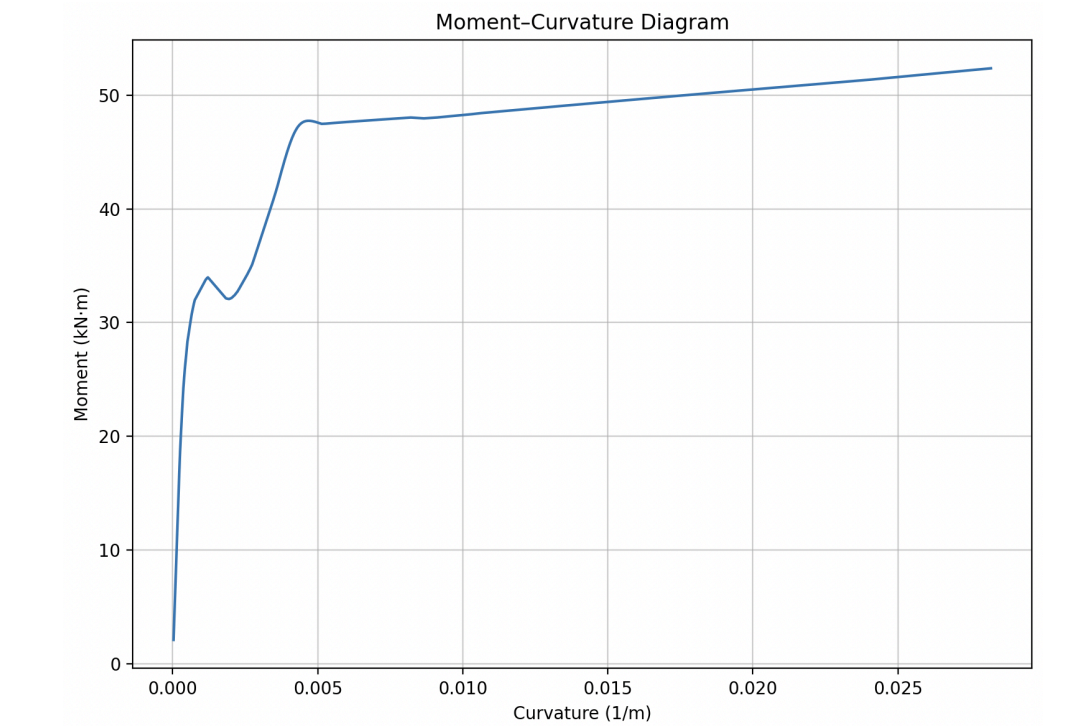


Fig 1. Moment Curvature (Monotonic Loading)

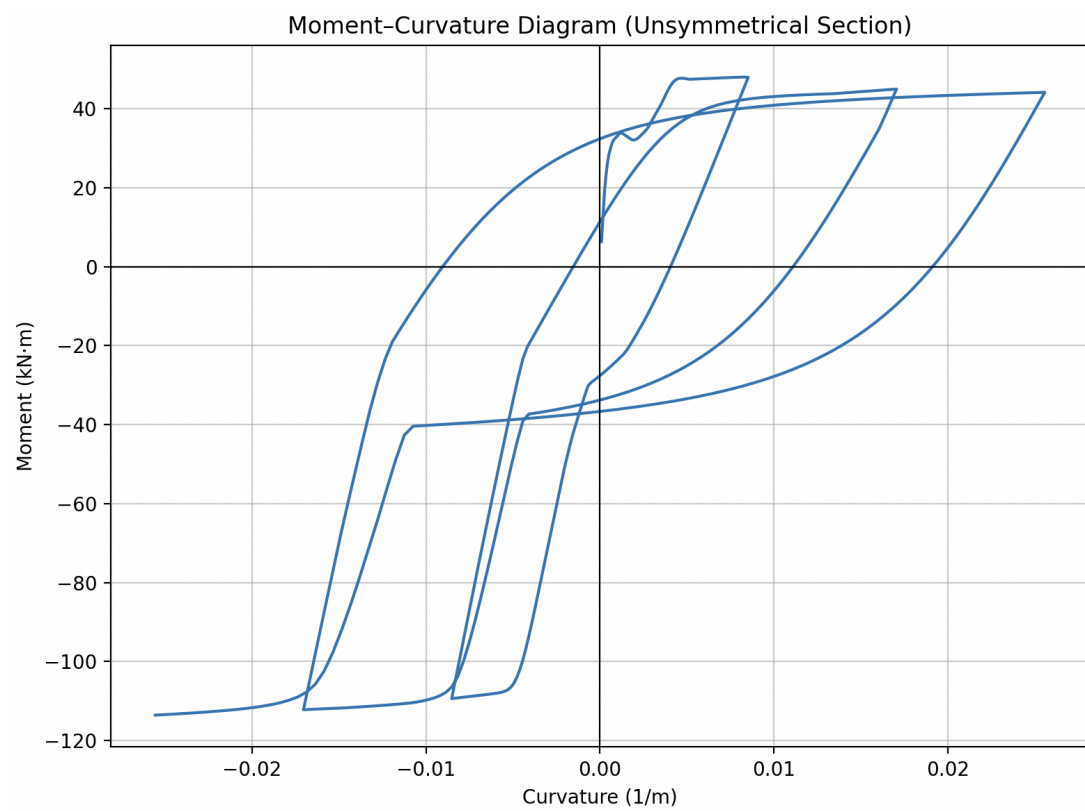


Fig 2. Moment Curvature (Cyclic Loading)

Cantilever Setup:

Flexible length: 1916 mm

Rigid joint length: 203 mm

Total length: 2.119 m

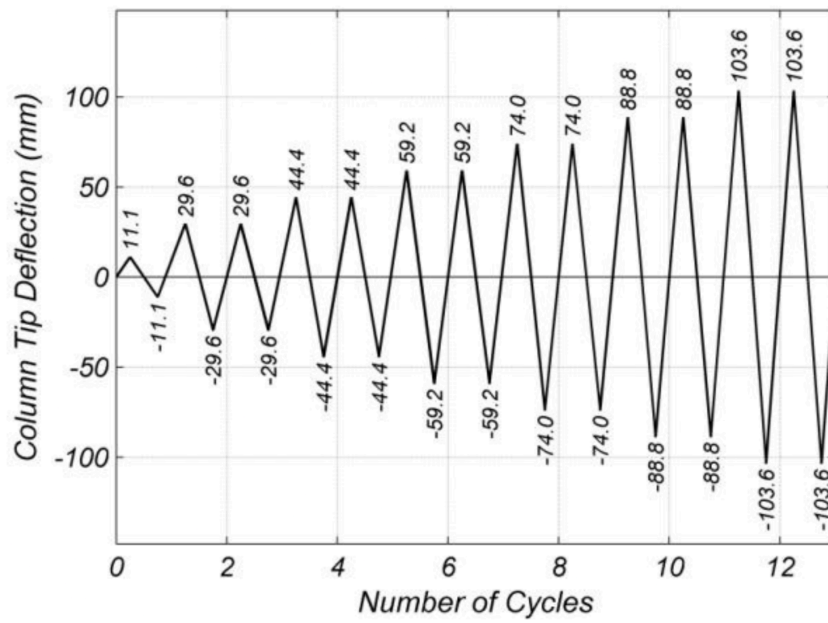


Fig 3. Imposed Cantilever Tip Deflection

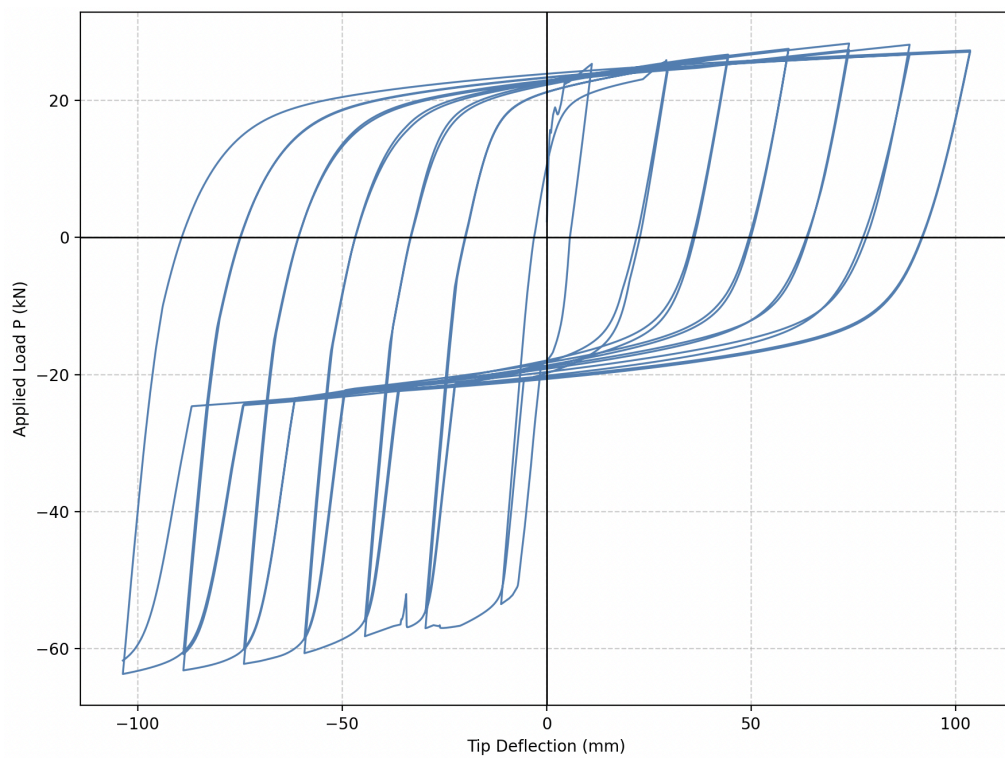


Fig 4. Load-Deflection Curve of Cantilever Assembly

Table 1. Comparison of Manual vs Numerical Results

Parameter	IS 456:2000 Nominal	OpenSees
Sagging Moment Capacity ($M_{nominal}^+$)	47.64 kNm	≈ 47 kNm (Figure 1)
Hogging Moment Capacity ($M_{nominal}^-$)	109.26 kNm	≈ 108 kNm (Figure 2)
Yield Curvature (ϕ_y)	4.23×10^{-3} 1/m	4.23×10^{-3} 1/m at $M^+ = 47$ kNm
Positive Cantilever Tip Load (P^+) at yield	22.48 kN	≈ 22.6 kN (Figure 4)
Negative Cantilever Tip Load (P^-) at yield	51.56 kN	≈ 51 kN (Figure 4)

Observations And Conclusion:

The numerical model results show very close agreement with the manual calculations performed in accordance with IS 456:2000, confirming the accuracy and reliability of the developed numerical models.

Effect of Section Asymmetry: Section A-A contains approximately 2.5 times more reinforcement at the top than at the bottom. As a result, the section exhibits a significantly higher negative moment capacity of 109.26 kNm (from nominal calculation) compared with its positive moment capacity of 47.64 kNm (from nominal calculation).

Effect of Gravity Loading: The applied gravity load of 55 kN on each beam generates a static negative moment at the joint face of ≈ 35.48 kNm ($55 \text{ kN} \times 0.645\text{m}$). This initial negative moment shifts the moment-curvature response. Consequently, under the negative cycle of the loading, the existing gravity-induced moment is amplified, leading to a larger negative curvature of approximately -0.20m^{-1} . In contrast, under the positive cycle of the loading, the applied moment must first overcome the gravity-induced negative moment before positive curvature can develop, resulting in a comparatively smaller positive curvature of approximately $+0.05\text{m}^{-1}$.

Global and Local Structural Behaviour: When the tip of the assembly is displaced to the right, the left beam experiences significantly higher negative bending due to the combined effect of the applied lateral load and the gravity-induced negative moment. In contrast, the right beam develops a smaller positive bending moment, as part of the applied moment is first required to overcome the existing gravity-induced negative moment. When the frame is displaced to the left, the roles of the two beams are reversed, with the right beam resisting the larger negative bending and the left beam resisting the smaller positive bending.

Although the individual beam responses are asymmetric because of the unequal reinforcement distribution and the presence of gravity loading, the interchange of beam roles during load reversal results in the same overall lateral resistance. Consequently, the structure exhibits symmetrical global behaviour despite asymmetrical local behaviour.